



The equation of a straight line

1 Match the keywords to their definitions. Write the correct letter in each box

a	constant
b	coefficient
c	gradient
d	y-intercept
e	origin
f	equation of a straight line

<i>b</i>	a numerical multiplier of a specific variable in a term
<i>c</i>	a measure of how steep a line is
<i>d</i>	the coordinate where a line crosses the y axis
<i>e</i>	the coordinate (0, 0)
<i>f</i>	a relationship which, when plotted, forms a straight line
<i>a</i>	a term that does not vary

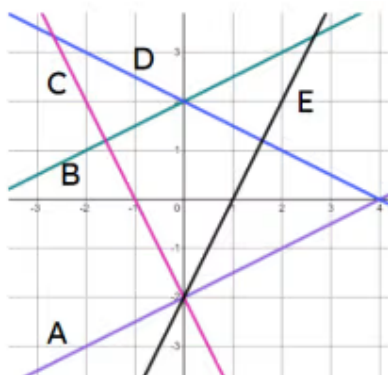
2 Which of these relationships are represented by this bar model? Tick **3** correct answers

<i>y</i>	5
<i>x</i>	

- ☐ $y - x = 5$
☐ $y + x = 5$
☒ $x - y = 5$
☒ $x = y + 5$
☒ $y = x - 5$



- 3 The graph shows 5 lines labelled A, B, C, D and E. Match the line to its gradient and y intercept. Write the correct letter in each box



a	A (purple)
b	B (green)
c	C (pink)
d	D (blue)
e	E (black)

e	gradient 2, y-intercept (0, - 2)
c	gradient - 2, y-intercept (0, - 2)
a	gradient $\frac{1}{2}$, y-intercept (0, - 2)
b	gradient $\frac{1}{2}$, y-intercept (0, 2)
d	gradient $-\frac{1}{2}$, y-intercept (0, 2)

- 4 Match the equations of lines to their key features. Write the correct letter in each box

a	$y = 5x - 4$
b	$y = 4 - 5x$
c	$y = -4x - 5$
d	$y = 4x + 5$
e	$y = 5x + 4$
f	$y = 5 - 4x$

a	gradient 5, y-intercept (0, - 4)
c	gradient - 4, y-intercept (0, - 5)
d	gradient 4, y-intercept (0, 5)
b	gradient - 5, y-intercept (0, 4)
e	gradient 5, y-intercept (0, 4)
f	gradient - 4, y-intercept (0, 5)



5 What are the key features of the graph with equation $y = 3(5 - 2x)$? Tick **1** correct answer

☐ gradient 15, y-intercept $(0, -6)$

☐ gradient 6, y-intercept $(0, -15)$

☐ gradient -2 , y-intercept $(0, 15)$

☐ gradient -2 , y-intercept $(0, 5)$

☒ gradient -6 , y-intercept $(0, 15)$

6 Which coordinates are on the line $2x - y = 5$? Tick **2** correct answers

☒ $(-2, -9)$

☐ $(-1, 3)$

☐ $(0, 5)$

☒ $(7, 9)$

☐ $(10, 5)$

